WEARABLE · SENSOR FUSION · HAPTIC THERAPY

CALMA

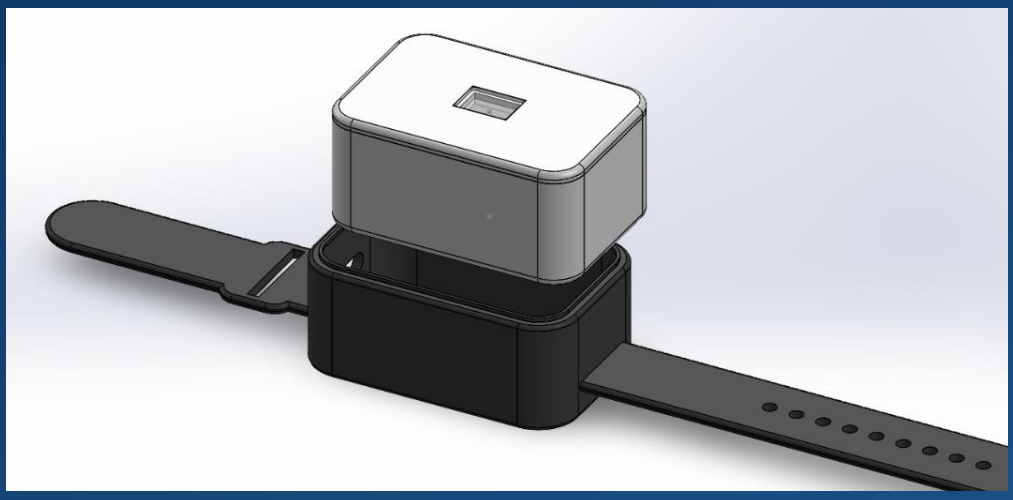
BAND

Wearable Stress-Awareness Band

Product Design Report

Prototype Design, CAD Model, Web Dashboard and Build Method

Predict → Detect → Calm



SolidWorks assembly of the CALMA prototype: housing, lid and silicone strap

4	75 × 50 × 30	4
sensing channels	mm design envelope	CAD parts, one assembly

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Subject	Product design of the CALMA wearable prototype
Scope	Industrial design, components, CAD model, web dashboard, prototype build method
Design tool	SolidWorks Premium 2025 SP4.1 (units: mm, g, s)
Date	October 2026

SECTION 1

1 Product Overview and Design Intent

CALMA is a wrist-worn wellness band that watches for early physiological signs of stress and responds with a gentle, hands-free intervention. It combines four sensing channels: optical pulse (PPG), electrodermal activity (EDA/GSR), motion (IMU) and skin temperature. The microcontroller fuses these signals, so that a raised pulse caused by walking or exercise is not mistaken for stress. When a stress-like pattern is detected at rest, the band delivers a breathing-synchronised haptic pattern and changes the colour of an RGB indicator. Readings are streamed over Bluetooth to the CALMA web dashboard.

This report documents the product design of the prototype: its physical form, the components housed inside it, the SolidWorks CAD model, the dashboard interface and the method used to build the prototype.

1.1 Design goals

- **Watch-like wearability.** A compact housing on top of the wrist with a soft strap, so the band can be worn for long periods.
- **Reliable skin contact.** The optical window and the two GSR electrodes face the skin, while the bulk of the electronics sits on the outer side.
- **Two-zone modular architecture.** Electronics (Zone A) and skin-contact sensors (Zone B) are separated to ease assembly, protection and comfort.
- **Protected electronics.** A rigid shell with a sealed lid and a USB-C slot, designed for 3D printing.
- **Clear feedback.** An RGB indicator and haptic motor on the band, plus a calm, readable dashboard on screen.

IMPORTANT

CALMA is a wellness and stress-awareness tool. It does not diagnose, treat or prevent any medical condition.

SECTION 2

2 Product Design Concept

The band follows the proportions of a thick sports watch. A rounded rectangular housing sits on the top of the wrist and a flexible strap wraps around it. The housing is split into two functional zones, shown below with the prototype concept visualisation.

2.1 Zone architecture

Zone	Location	Contents	Design rationale
Zone A	Upper housing, on top of the wrist	ESP32 board, Li-Po battery, TP4056 charger, MPU6050 IMU, vibration motor, RGB LED	These parts need no skin contact. Placing them on top protects them and balances the band.
Zone B	Lower face, against the skin	MAX30102 optical window, two GSR electrodes, MAX30205 temperature sensor	Skin-contact sensors need a stable, light-tight and clean interface to the wrist.

Table 1. *Functional zones of the housing.*

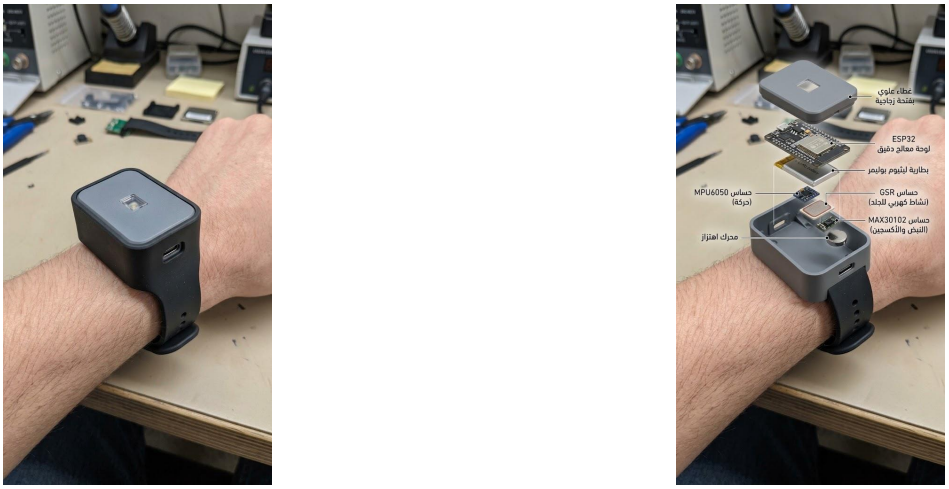


Figure 1. *Product design concept: the assembled CALMA band on the wrist (left) and an annotated exploded view of its internal components (right).*

2.2 Key dimensions

The initial design envelope was set at 75 mm long, 50 mm wide and 30 mm high, with a 25 mm strap width and a 2.0 mm wall thickness, which leaves room for the battery and ESP32 stack. The part-level specification used for the CAD model is given below.

Part	Length (mm)	Width (mm)	Height (mm)	Material	Fixing / clearance
Main housing	60	35	18	PETG / ABS	M2 screws at 4 corners; 1.0 mm internal clearance
Top lid	60	35	2	PETG / ABS	Screws / press-fit; 0.2 mm edge clearance
Lower sensor plate	40	25	3	PLA / resin	Press-fit inside housing; 0.5 mm clearance
MAX30102 cavity	16	16	2	Clear lens	Glued or press-fit; 0.5 mm lens clearance
GSR electrode seats	10 (dia.)	10 (dia.)	2	Stainless steel	Raised metal nuts
Wrist strap	220	22	2	TPU / silicone	Mounting slots integrated in the housing

Table 2. *Part-level mechanical specification (from the engineering report).*

2.3 Key design features

- **Side slot for charging.** A slotted opening in the housing wall gives access to the USB-C port without removing the lid.
- **Sensor face.** The base has a central rectangular window for the optical sensor, with a round opening on each side for the GSR electrodes, so all skin-contact sensors share one flat surface.
- **Soft edges.** Filleted corners and a chamfered rim keep the housing comfortable against the wrist and give it a watch-like look.
- **Stepped lid.** The lid edge steps into the rim of the housing, which locates it accurately and helps keep sweat out.
- **Flexible strap with cradle.** The silicone strap holds the housing in a pocket and has nine adjustment holes for different wrist sizes.

SECTION 3

3 Internal Components

The table summarises every component placed inside the band. Thumbnails are taken from the engineering report; the MAX30205 image is cropped from the concept visualisation.

Image	Component	Model	Function	Size (mm)	Zone
	Microcontroller	ESP32-WROOM-32	Processing, sensor fusion and Bluetooth	52 × 28	A, upper housing
	Pulse sensor	MAX30102	Optical heart-rate (PPG) measurement	15 × 15	B, skin side
	Sweat sensor	Grove GSR module	Skin conductance (EDA/GSR) via two electrodes	24 × 20	B, skin side
	Motion sensor	MPU6050	Accelerometer and gyroscope; motion rejection	20 × 15	A, upper housing
	Temperature sensor	MAX30205	Skin temperature as a contextual metric	15 × 15	B, skin side
	Vibration motor	10 mm coin motor	Haptic breathing-sync pattern	10 × 3	A, inner wall
	Status indicator	WS2812B RGB LED	Visual device and alert status	5 × 5	A, outer face
	Battery	Li-Po 3.7 V, 500 mAh	Portable power supply	35 × 25 × 5	A, upper housing
	Charger	TP4056 with protection	USB-C charging and battery protection	25 × 19	A, upper housing
	Motor driver	2N2222 + 1N4007	Transistor switch and flyback diode for the motor	Very small	A, main board

Table 3. *Component list with thumbnails.*

DESIGN NOTE

The vibration motor draws more current than an ESP32 pin can supply, so it is driven through a 2N2222 transistor with a 1N4007 flyback diode across the motor. It is mounted on the inner wall of the housing so the vibration is transferred to the shell and to the wrist.

SECTION 4

4 CAD Model (SolidWorks)

The prototype was modelled in SolidWorks Premium 2025 SP4.1. It consists of four parts, main_box, top_lid, lens and strap, combined in a single assembly (Assem1) with mates between the components. The sections below describe each part, followed by the assembled product.

Part	Material in CAD	Main features in the feature tree
main_box	ABS	Boss-Extrude1, Shell1, Cut-Extrude1, Cut-Extrude2, Cut-Extrude8, Fillet1, Chamfer1
top_lid	ABS	Boss-Extrude1, Boss-Extrude2, Cut-Extrude1, Fillet1, Chamfer1, Cut-Extrude2
lens	Acrylic (medium-high impact)	Boss-Extrude1, Boss-Extrude2 (stepped transparent part)
strap	Silicone rubber	Boss-Extrude (several), Extrude-Thin2 and Extrude-Thin4 for the walls, Cut-Extrude for the holes, Fillet4 to Fillet12

Table 4. *Parts of the CAD model. The materials shown are those assigned in SolidWorks for visualisation.*

4.1 Main housing (main_box)

The housing is a rounded rectangular box made by extruding a sketch and shelling it into a hollow shell. Its walls are filleted and a chamfer softens the rim where the lid closes. A slot in one side wall provides access for the USB-C charging port. The base carries a central rectangular window for the optical sensor with a round opening on each side for the GSR electrodes.

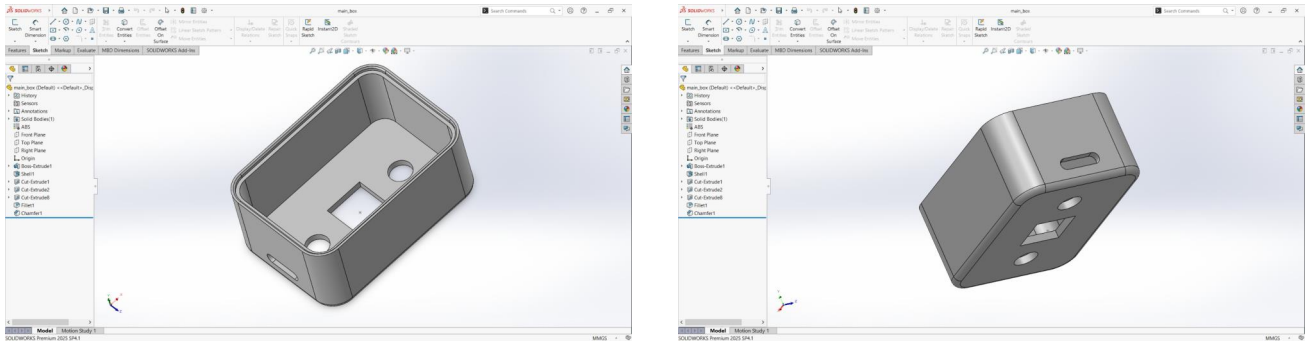


Figure 2. *main_box: interior view with the internal shelf and openings (left) and the underside with the sensor window and electrode holes (right).*

4.2 Top lid (top_lid)

The lid is a flat plate with rounded corners and a stepped edge that fits into the rim of the housing with a small clearance. A rectangular opening in its centre allows the indicator to be seen.

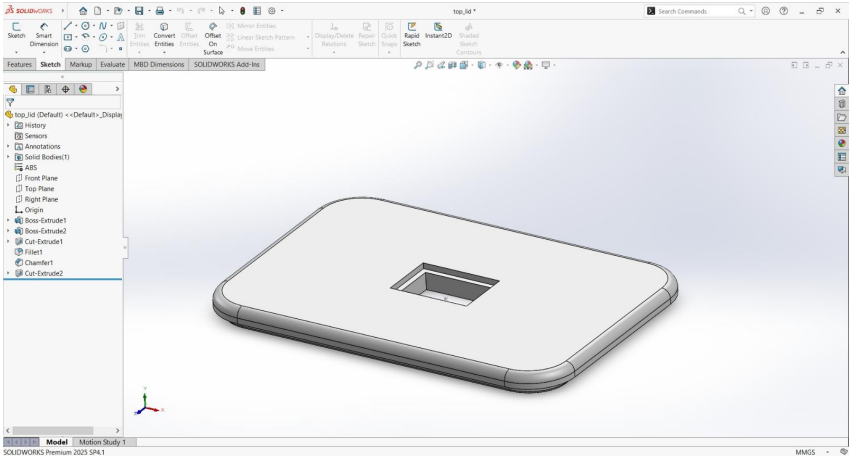


Figure 3. *top_lid: stepped edge, rounded corners and a central rectangular opening.*

4.3 Lens

The lens is a two-step transparent part with rounded corners, modelled in acrylic. Its narrow lower step is designed to seat in a window while the wider upper step forms a flange, giving a clear, sealed path for the optical sensor.

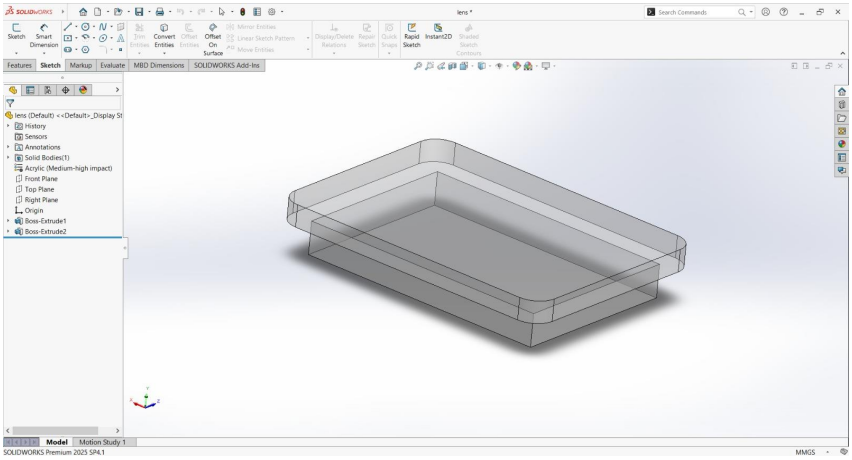


Figure 4. *lens: stepped transparent part (shown with transparency enabled).*

4.4 Wrist strap

The strap is a single flexible part modelled in silicone rubber. A rectangular cradle in the middle holds the housing. One end carries a retaining loop with a slot, and the other end has nine adjustment holes, so the band can be sized to different wrists.

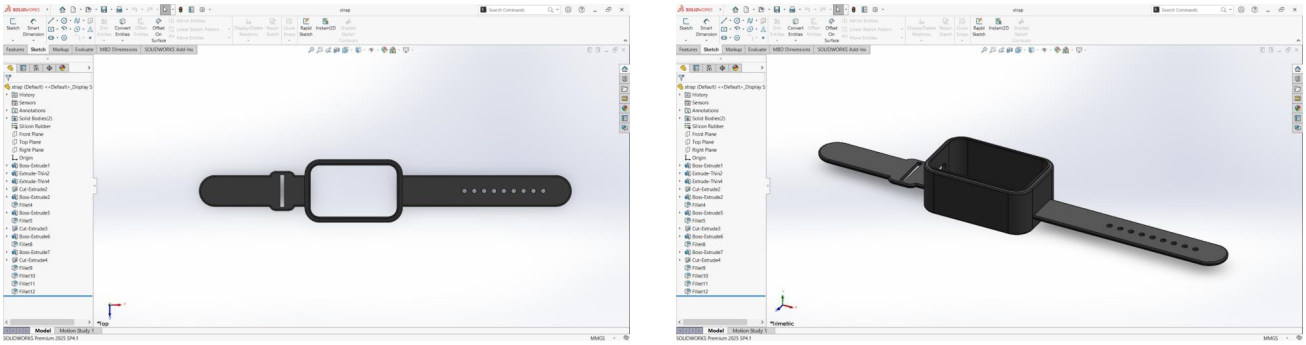


Figure 5. *strap: top view (left) and trimetric view (right).*

4.5 Assembly

In the assembly the housing sits in the strap cradle and the lid closes the top. The views below show the components during assembly in SolidWorks, together with the feature tree of each part.

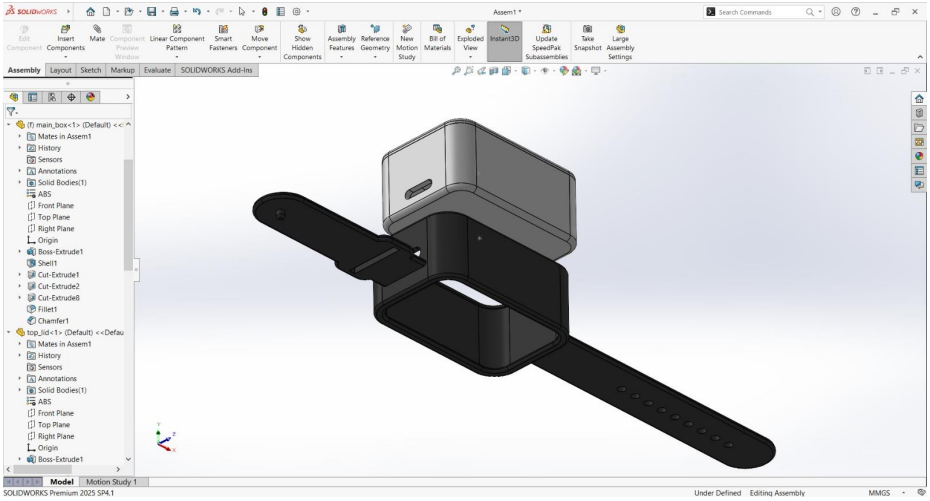


Figure 6. *Assembly view (Assem1) showing main_box and top_lid above the strap, with the main_box feature tree.*

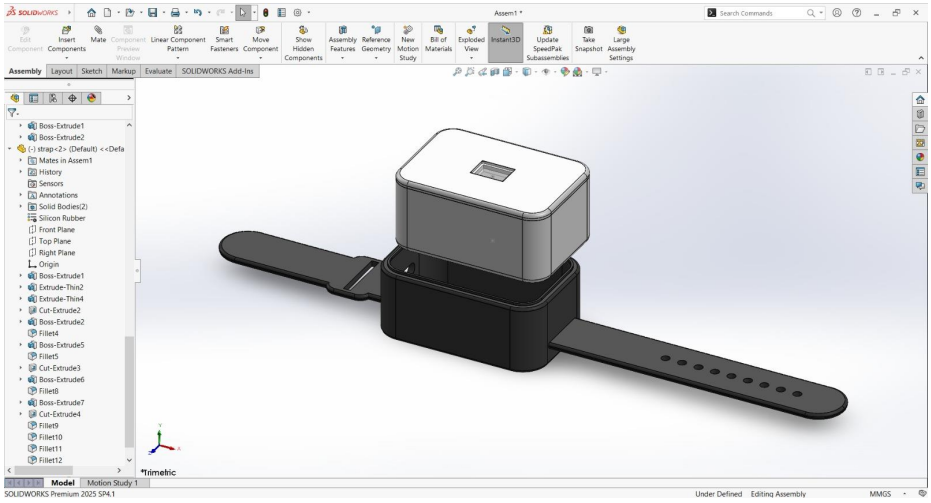


Figure 7. *Assembly view with the strap component and its feature tree; the housing and lid sit above the strap cradle.*

SECTION 5

5 Web Dashboard Design

The band is paired with a web dashboard that turns the sensor data into calm, readable information. The interface uses a dark theme with a teal accent. Large status cards and soft rounded panels keep the screen quiet, which suits a product meant to reduce stress. Every screen carries the wellness disclaimer in its footer.

Sensing	→	Processing	→	Decision	→	Intervention	→	Dashboard
PPG, GSR, IMU, temperature		ESP32 filtering and peak detection		Compare with personal baseline		Haptic pattern and RGB LED		Bluetooth to web app

Data flow from the band to the dashboard.

5.1 Interface structure

- **Navigation.** A left sidebar with Overview, Live Monitoring, Calm Sessions, Episodes, Analytics, Insights, AI Assistant, Device and Settings.
- **Status bar.** Connection status, notifications and the user profile at the top; band connection and battery level at the bottom of the sidebar.
- **Visual language.** Rounded cards, one accent colour, large numbers for key values and small supporting labels.
- **Personal baseline.** Charts show a dashed baseline so each reading is judged against the user, not a fixed norm.

5.2 Overview

The landing screen answers one question: how am I doing right now? A large wellness status card shows the current state (Stable) and a stress indicator ring (24). Four tiles show heart rate, skin conductance, skin temperature and motion. Below, a pattern chart with 1H, 6H, 12H and 24H ranges, a Quick Calm Session button and the most recent episode complete the view.

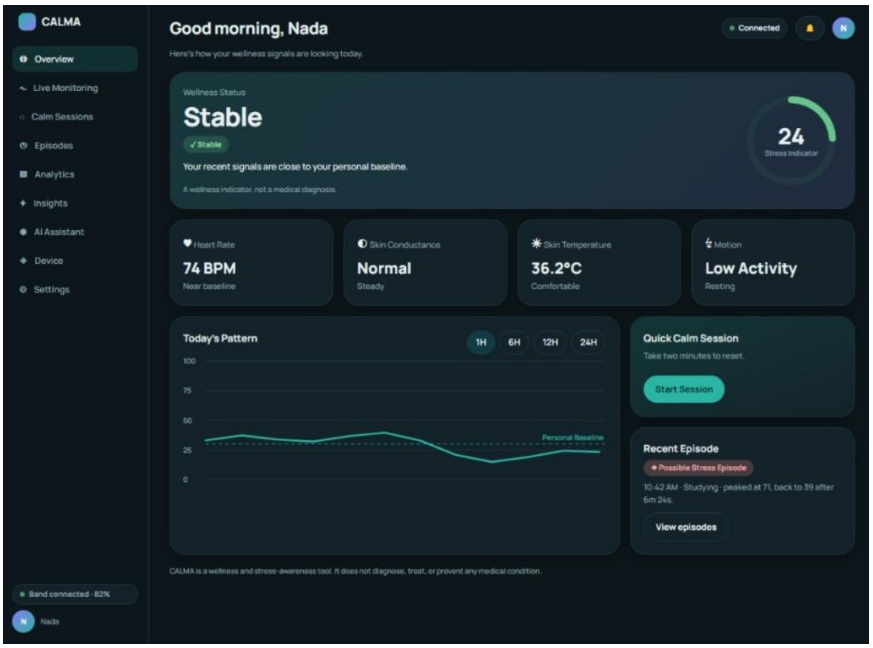


Figure 8. *CALMA Overview screen.*

5.3 Live Monitoring

This screen shows real-time data. A LIVE badge and a status chip (Elevated) sit above four metric cards: stress indicator (42 / 100), heart rate (75 BPM), GSR (3.2 µS) and temperature (36.3 °C). The chart can be set to 1 minute, 5 minutes, 15 minutes, 30 minutes or 1 hour, and individual signals (heart rate, GSR, temperature, motion, stress indicator) can be switched on or off.



Figure 9. *CALMA Live Monitoring screen.*

5.4 Analytics

Analytics summarises the week: number of episodes (5), average duration (5 min 12 s), longest episode (6 min 24 s) and the most active period (2 to 4 PM). A weekly indicator chart, a recovery trend, a time-of-day heatmap and a context comparison (studying, working, exercising, sleeping, other) help users see when and why stress appears.

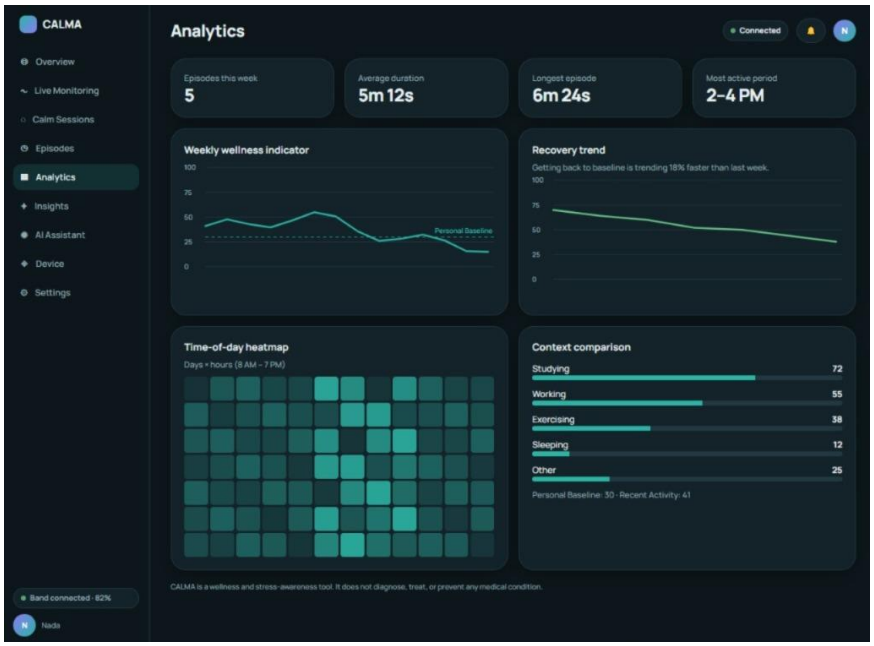


Figure 10. *CALMA Analytics screen.*

5.5 Device

The Device screen shows the status of the band: battery level (82%), Bluetooth connection, firmware version (v1.0.4) and whether the heart rate, skin conductance, temperature and motion sensors are active. Three actions are provided: Test Sensors, Calibrate Device and Reconnect.

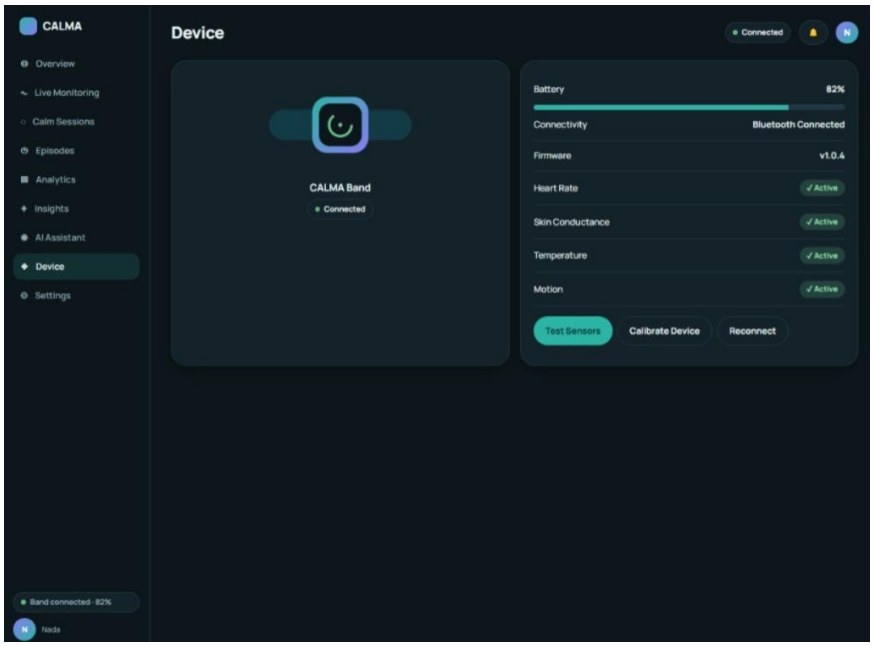


Figure 11. *CALMA Device screen.*

5.6 AI Assistant

The AI Assistant lets users ask about their own data in plain language. In the example, the user asks why the indicator rose today. The assistant explains that heart rate and skin conductance were above the personal baseline around 2:15 PM, notes the period was tagged as Studying, and offers a 3-minute calming session. Suggested prompts (patterns, recovery trend, calming session, latest episode) make the first step easy.

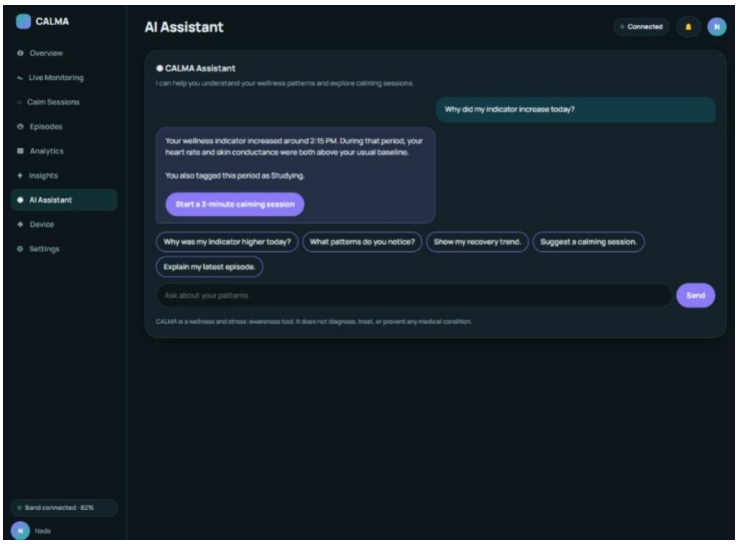


Figure 12. *CALMA AI Assistant screen.*

SECTION 6

6 Prototype Implementation Method

The prototype is built in stages. Each electronic block is proven on its own before the parts are placed in the printed housing, because the final dimensions depend on the real size of the assembled board.

6.1 Build phases

Phase	Activity	Description
1	Purchase and preparation	Buy the development board and sensors locally.
2	Bench testing	Connect each sensor alone on a breadboard and confirm its readings on the serial monitor.
3	Sensor integration	Connect the I2C devices together on the ESP32 and check that their addresses do not clash.
4	Motor driver circuit	Solder the transistor, diode and vibration motor, then test the PWM commands.
5	Firmware	Write the signal-processing code that fuses the readings and applies motion rejection.
6	3D design and printing	Design the housing and print it in ABS or PETG, with the strap in flexible TPU.
7	Final assembly	Move the circuit from the breadboard to a small perfboard to reduce its size.
8	Calibration	Test with volunteers at rest and under simulated stress to tune the thresholds.

Table 5. *Prototype build phases.*

6.2 Electrical connections

The three I2C sensors share one bus and are told apart by their addresses. The remaining parts use dedicated pins.

Component	Signal	ESP32 pin	Note
I2C bus	SDA	GPIO 21	Shared by MAX30102, MPU6050 and MAX30205
I2C bus	SCL	GPIO 22	Shared by MAX30102, MPU6050 and MAX30205
GSR sensor	Analog out	GPIO 34 (ADC1)	ADC1 keeps working while Bluetooth is active
Motor driver	PWM (base)	GPIO 25	Through a 1 kΩ resistor to the 2N2222 base
RGB LED	Data in	GPIO 26	330 Ω resistor protects the pin

Table 6. *Proposed ESP32 pin assignment.*

6.3 3D printing and materials

- Rigid parts that carry the electronics are printed in PETG or ABS with a wall thickness of about 2.0 mm.
- The wrist strap is printed in flexible TPU, so the sensors are held against the skin without discomfort.
- The lid is fixed with M2 screws, and clearances of 0.2 to 1.0 mm are kept on the mating faces.
- The GSR electrodes are stainless steel (or gold-plated metal), and the housing joints are sealed to keep sweat out.

6.4 Assembly sequence

1. Solder the core circuit (ESP32, resistors, transistor) on a small perfboard sized to the housing.
2. Fit the GSR electrodes and the PPG window in the lower base and glue them in place.
3. Wire the lower sensors (MAX30102, GSR, MAX30205) to the main board.
4. Fix the vibration motor and the IMU to the inner walls of the upper housing so vibration is transmitted.
5. Mount the RGB LED facing the opening in the lid.
6. Solder the battery to the TP4056 charger and connect its output to the board power input.
7. Place the battery and circuit inside the housing.
8. Close the housing with the screws, taking care not to pinch the wires.
9. Attach the TPU strap.

6.5 Verification

Test	Method	Expected result
Electrical	Measure voltage with a multimeter while the motor runs	Stable 3.3 V at the sensors
Pulse (PPG)	Compare with a commercial pulse oximeter at rest	Difference within ±5 beats
Sweat (GSR)	Take a sudden deep breath and watch the signal	Response within 2 to 3 seconds
Motion (IMU)	Shake the band during a pulse reading	Pulse reading is paused; no false alarm
Vibration	Run different PWM patterns	Clear vibration with no ESP32 reset
Contact	Loosen the strap	Poor-contact warning is shown

Table 7. *Prototype verification plan.*

6.6 Design considerations for the next iteration

- Use Bluetooth Low Energy rather than Wi-Fi to extend battery life, and measure the real current draw before fixing the 500 mAh battery.
- Extend motion rejection to the GSR channel, and add voltage regulation for a stable sensor supply.
- Replace the perfboard with a custom, smaller PCB and add machine-learning models for stress detection.
- Fix the final housing dimensions only after the assembled board is measured.

DISCLAIMER

CALMA is a guidance and wellness tool. It is not an approved medical diagnostic device.